

Axiom Governance for an AI-Generated Formal Library: Taxonomy, Redemption Methodology, and Honest Proof-Rate Metrics

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baseline, landed 2026-08-21)

Abstract

AI-assisted formalization scales content faster than the human communities of practice that traditionally guarantee it, and the debt accumulates in three specific failure modes: axioms used as deferred proofs, proof-rate metrics that drift with the measuring caliber, and string-level scans that mistake comments for placeholders. We document the axiom-governance program of TOE-SYLVA, a Lean 4 (+ Agda) library built on a pinned mathlib4 revision and containing [⟨119859⟩](#) machine-generated batch modules alongside handcrafted developments. The program comprises: (i) a four-class axiom taxonomy (primitive, definitional, schema, placeholder) with a machine-readable registry of [⟨448⟩](#) axioms; (ii) a redemption methodology executed across six sweeps, whose [⟨196⟩](#) closures we re-classify into four semantic grades — vacuous placeholder cleanup, trust transfer ($P \rightarrow P$ conditionalization), definitional reconstruction, and substantive proof — of which only [⟨13⟩](#) carry non-trivial proof content; (iii) an honest proof-rate metric that counts code-level `sorry` with comments stripped, replacing a caliber history that drifted from [⟨88.08%⟩](#) to [⟨100%⟩](#) and then collapsed to [⟨3.91%⟩](#) once axioms were no longer ignored; and (iv) a four-tier external claim layer (THEOREM / THEOREM* / CLAIM / CONJECTURE) with falsifiability criteria. Two incidents in which the governance loop caught defects in its own metrics — a [⟨30.2%⟩](#) “sorry contamination” alarm that reversed to zero under a code-level caliber, and a 25× disagreement between two dashboards measuring the same library — are elevated into the methodological thesis of the paper: *every proof-library metric must be bound to an explicitly declared caliber level (string / code / CI-pattern)*, and the binding must be machine-checkable. We close with a full lifecycle case study — a CLAIM→THEOREM upgrade along a Chern–Simons dependency chain with mathlib-delegation honesty labels — and the quarantine of the single genuinely contaminated module in the batch corpus.

Draft-skeleton status (D3 compilation pilot). Numbers in red angle brackets $\langle \dots \rangle$ are declared calibers retained for traceability. The P0 canonical-number reconciliation (D1 review sec 1.4-1) has landed (`framework/p0_reconciliation_report.md`, 2026-08-21): all C6-related figures — sweep accounting (196), semantic-grade decomposition, registry (448) vs. live-lean (253) calibers — are canonical and quotable; the remaining bracketed values are verified v7.98 baselines, each annotated with its repository source in the L^AT_EX source comments. The full conflict inventory (C1–C6) that motivated this discipline is tabulated in Sec. 3.6.

1 Introduction

1.1 Setting: a library that scaled faster than its governance

TOE-SYLVA is a formalization program whose Lean 4 side is built strictly on top of a pinned mathlib4 revision (commit 8a178386 at v7.98; toolchain Lean 4.29.0) and contains two very different kinds of content. The first is a set of handcrafted developments targeting physics-adjacent

mathematics (Chern–Simons theory, topological invariants, gauge structures, algorithmic complexity). The second is a machine-generated *batch corpus* of [119859](#) modules following the naming pattern `SYLVA_Proven*R*M*`, produced by a generator whose manifest ([119831](#) entries) was verified by SHA-256 sampling ([500](#) draws, zero mismatches at v7.98). Alongside it, an Agda side carries [7](#) files.

This scale was reached with AI assistance, and it inherited the three failure modes that motivate the present paper. They are not hypothetical: each was caught, documented, and (partially) remediated inside the repository, and the remediation records are themselves the primary material of this article.

1.2 Three failure modes of scaled formalization

F1: axioms as deferred proofs. At its v7.98 baseline the registry lists [448](#) axiom declarations across [116](#) non-batch files. A substantial fraction of these were not mathematical primitives the framework genuinely needs, but *placeholders*: statements the generation pipeline could not prove at production time and parked as `axiom` so that downstream files would elaborate. The registry’s `placeholder` class contains [42](#) such entries that a naive name-based grep over `axiom` declarations would not flag as problematic at all.

F2: proof-rate caliber drift. The library’s headline “proof rate” was reported at different times as [88.08%](#), then [99.5%](#), then [100%](#), and — after the honest metric of Sec. 2.3 was adopted — as [3.91%](#) for the handcrafted corpus. None of these numbers was fabricated: each was correct under the caliber in which it was computed (sorry-count only; axioms ignored; per-module normalization). The $25\times$ spread is a property of the *calibers*, not of the computations — which is precisely why caliber discipline must become an explicit, first-class part of the methodology rather than an implicit convention.

F3: string-level false positives. A routine string search reported that [36199](#) of the [119859](#) batch modules (“[30.2%](#)”) *contain sorry*. A full-population audit under a comment-stripping code-level caliber reversed the finding: [36198](#) of those hits were the literal text of a *negation* header comment (Actual Lean 4 proofs --- no sorry, no True := trivial), the code-level count is zero, and exactly one file (`SYLVA_ProvenLayer2.lean`) constitutes a genuine, if ironic, contamination case: a theorem *named* `sylvia_zero_sorry_guarantee`, proved by `True := trivial`, inside a file whose header claims that it contains no trivial placeholders (Sec. 3.4).

1.3 The governance program

The remediation program has four pillars, each mapping to one section of the methodology:

1. **Taxonomy and registry** (Sec. 2.1): every axiom is classified into one of four classes and recorded in a machine-readable JSON registry with per-entry provenance.
2. **Redemption methodology** (Sec. 2.2): axioms are discharged in recorded sweeps, and every closure is re-classified into one of four *semantic grades* — including the unflattering ones (vacuous discharge, trust transfer) — so that the redemption statistics cannot inflate themselves.
3. **Honest proof-rate metric with declared calibers** (Sec. 2.3): a metric that strips comments before counting `sorry`, zeroes out files that declare axioms or postulates, and is always reported together with its caliber level.

4. **External claim layer** (Sec. 2.4): a four-tier labeling (THEOREM / THEOREM* / CLAIM / CONJECTURE) with falsifiability criteria per claim, separating the internal formal status of a lemma from the external epistemic status of a physics claim.

1.4 Contributions

- C1** A four-class axiom taxonomy and machine-readable registry for a mixed human/AI formal library ([448](#)) entries at v7.98 baseline).
- C2** A redemption methodology with honest semantic grading: across six sweeps, [196](#) axioms were closed (clearing the [142](#)-entry **definitional** class entirely), of which only [13](#) are substantive proofs; we argue that reporting this decomposition — rather than a single redemption count — is the correct epistemic unit of account.
- C3** A proof-rate metric bound to an explicit caliber level, together with a three-tier caliber standard (string / code / CI-pattern) adopted as repository policy.
- C4** Two documented cases of the governance loop catching defects in its own metrics (F2, F3), argued to be the strongest available evidence that the methodology is doing work.
- C5** A full lifecycle case study — CLAIM→THEOREM upgrade along a Chern–Simons dependency chain with explicit mathlib-delegation labels (Sec. 3.5) — and the quarantine protocol for the single contaminated batch module (Sec. 3.4).

1.5 Transparency statement

This library was produced with AI assistance, and so was this paper’s skeleton. The repository discipline it describes — registration first, boundaries explicit, no silent deletions — is applied to the paper itself: every number is a labeled baseline pending a canonical reconciliation (P0), every claim about the library cites the internal audit report it rests on, and the draft status of each section is marked. The supplementary build artifacts (compile log, PDF) accompany the skeleton under `framework/paper/`.

2 Methodology

2.1 Axiom taxonomy and registry

Every axiom declaration in the non-batch corpus is registered in a machine-readable JSON registry (`axiom_registry.json`, schema v2.0) with one of four classes:

Class	Semantics	Disposition
primitive	genuinely assumed primitive	keep, declare
definitional	restatement of a definition	discharge by <code>rfl/norm_num/simp</code>
schema	instance of a general transfer pattern	conditionalize ($P \rightarrow P$)
placeholder	deferred proof parked as axiom	full replacement or explicit debt

At the v7.98 baseline the registry contains [448](#) entries over [116](#) non-batch files; the **placeholder** class alone holds [42](#) entries. The registry is the single point of truth for redemption accounting (Sec. 2.2); its per-entry provenance (file, line, sweep that closed it, closing transformation) is what makes the semantic grading of Sec. 2.2 auditable rather than assertive.

2.2 Redemption methodology and semantic grading

Axioms are discharged in recorded *sweeps* (six at the reconciled working tree: sweep1–sweep6), each producing a per-entry report. The methodological commitment of this paper is that a sweep’s output must be graded by *semantic content added*, not by closure count. Four grades:

Grade	Content added	Typical transformation	Count
G1 vacuous	none	<code>axiom T : True → theorem T : True := trivial</code>	⟨50⟩
G2 trust transfer	none (hypothesis)	<code>axiom P → theorem (h : P) : P := h</code>	⟨90⟩
G3 definitional	definitional equality	<code>data axiom → noncomputable def; rfl</code> rewrites	⟨41⟩
G4 substantive	real proof content	arithmetic / existence proofs	⟨13⟩
<i>excluded from redemption</i> : axiom bundling (kept, declared)			⟨2⟩

Two design points deserve emphasis. First, *conditionalization* (G2) is honest but content-free: it converts an axiom into a theorem whose hypothesis is the axiom’s statement, preserving downstream elaboration while making the debt visible at every use site. Second, the grade distribution itself is the deliverable: only [⟨13⟩](#) of [⟨196⟩](#) closures are G4. Reporting only “[⟨196⟩](#) axioms redeemed” would be the exact failure mode (F2) this methodology exists to prevent.

2.3 The honest proof-rate metric and caliber discipline

The repository’s normative definition (`proof_status.md` sec 5):

true proof rate = (theorems derived from axioms and known theorems with zero `postulate`) / (total claimed theorems).

Operationally, two calibers coexist and both are legitimate when declared:

- `[code]`: `sorry` is counted after stripping comments; files declaring `axiom/postulate` contribute zero true proofs. Reference implementation: `count_sorry_no_comments` in `compute_proof_rate`. — the single normative implementation, not to be reimplemented ad hoc.
- `[independent]`: additionally excludes trivially-patterned proofs and axiom-dependent theorems, yielding the strict “independent derivation” rate.

The three-tier caliber standard adopted as policy after the incident of Sec. 3.3 is:

Tier	Meaning	Admissible for
<code>[string]</code>	raw substring match, comments included	nothing (quarantine for humans)
<code>[code]</code>	comment-stripped, code-level pattern	contamination judgment, CI
<code>[CI-pattern]</code>	anchored CI grep with comment exclusion	gates, dashboards

The rule: *a metric is not a number; it is a (number, caliber) pair*, and any headline number without a declared caliber is treated as a defect (cf. C3, C4 in Sec. 3.6).

2.4 The external claim layer

Internal formal status and external epistemic status are separated by a four-tier label, assigned per claim and registered centrally (`proof_status.md`):

Tier	Standard
THEOREM	formal proof machine-checked in-repo, core free of postulate
THEOREM*	mathematical body proven, relying on known theorems not re-verified in-framework (reference integrity confirmed)
CLAIM	reasoning / numerics / citation support, no independent formalization or review
CONJECTURE	speculative, logically coherent, falsifiable

Every CLAIM and CONJECTURE carries a *falsifiability criterion* (e.g., for the fine-structure coincidence $\alpha^{-1} \approx n_{CS} = 137$: a first-principles computation of α^{-1} to ± 0.01 independent of the loop-integer self-tuning parameters). The registered entries at v7.98 span FLT and Langlands (THEOREM*, via mathlib), CHSH and the Dedekind unique-factorization delegation (THEOREM), the stratified Chern-number identity (THEOREM, with `#print axioms` reduced to the standard trio `propext`, `Classical.choice`, `Quot.sound`), and the Standard-Model gauge-group, Chern–Simons level, and spectral action claims (CLAIM, with explicit gap chains).

2.5 CI discipline and the reproducibility envelope

The honesty layer of the repository is machine-checked (`verify_honest_repo.py --ci`: forbidden-term scan with deletion-context allowance, required honesty markers, deleted-claims ledger with > 10 entries, BibTeX integrity, new-paper presence). Its *boundary is stated explicitly*: the CI checks the honesty layer, not Lean proof content — conflation of the two is itself a caliber error. The Lean side is pinned: mathlib4 at commit `8a178386`, toolchain Lean 4.29.0, with independent compile logs recording `#print axioms` output. The batch generator is reproducible from its manifest (`<119831>` entries, SHA-256 sampling `<500>`, zero mismatches at v7.98).

3 Results

3.1 Registry and redemption outcomes

Across six sweeps the program closed `<196>` registered axioms (`<30>+<44>+<31>+<29>+<30>+<32>`), clearing the **definitional** class (`<142>` entries) and the **placeholder** class (`<42>`) entirely; live **axiom** declarations in `.lean` sources fell from `<445>` at the registry baseline (v7.94) to `<253>`, exactly matching the registry residual (primitive `<239>` + schema `<14>`). The semantic-grade decomposition is the headline result, and it is deliberately unflattering: of `<196>` closures, `<50>` are vacuous (G1), `<90>` trust transfers (G2), `<41>` definitional reconstructions (G3), and `<13>` substantive proofs (G4) — `<6.6%>` of closures, or about `<2.9%>` of the registered axiom population.

3.2 Proof-rate caliber history

The same library, measured at different times with different implicit calibers:

Reported	Caliber (implicit at the time)	Status
<code><88.08%></code>	sorry-count based	superseded
<code><99.5%></code> / <code><99.9%></code>	sorry-count, per-module	superseded
<code><100%></code>	(<code>#sorry = 0</code>) / (<code>#modules</code>); axioms ignored	superseded
<code><3.91%></code>	[<code>independent</code>]: excludes trivial patterns, sorry, axiom-dependent	current (2026-08-11)
<code><99.79%></code>	[<code>code</code>]: theorems – sorry, file-level axiom zeroing	current (parallel file)

The two “current” rows disagree by $25\times$. This is not an embarrassment to be patched quietly but the central exhibit of the paper: both numbers are arithmetically correct under their declared calibers, and the conflict inventory (Sec. 3.6) tracks its resolution as a first-class P0 task.

3.3 The 30.2% sorry incident and its reversal

A string-caliber scan reported $\langle 36199 \rangle$ of $\langle 119859 \rangle$ batch files ($\langle 30.2\% \rangle$) as “containing sorry”. A full-population audit (no sampling shortcut; every hit classified) reversed the finding:

Class of hit	Files	Nature
header negation comment	$\langle 36198 \rangle$ (99.997%)	Actual Lean 4 proofs --- no sorry, no <code>True := trivial</code> (a <i>denial</i> , matched as a substring)
code-level sorry	$\langle 0 \rangle$	—
genuine contamination	$\langle 1 \rangle$ (0.003%)	<code>SYLVA_ProvenLayer2.lean</code>

The reversal had an operational consequence beyond bookkeeping: the draft TrivialBench sampling protocol (A-2) filtered modules by the string caliber, which would have silently discarded $\langle 30.2\% \rangle$ of *qualified* corpus (leaving, e.g., only $\langle 14\% \rangle$ of Analysis -domain modules). The protocol clause has been revised to the code-level caliber with an explicit quarantine exception for the single genuine case.

3.4 Quarantine case: `SYLVA_ProvenLayer2.lean`

The one genuine case concentrates all three irony modes of AI generation. The file’s header asserts it contains “10 theorems with ACTUAL PROOFS. No `True := trivial` placeholders — all proofs are real.” In fact $\langle 8 \rangle$ of its $\langle 10 \rangle$ theorems are `True := trivial`; one of the two remaining is named `sylva_zero_sorry_guarantee` — the zero-sorry guarantee, proved by `trivial` — which made the file the only non-declaration string-caliber **sorry** hit in the entire batch corpus. Under the quarantine protocol the file was moved out of the batch corpus into an isolation directory with a machine-readable manifest (origin path, SHA-256, reason list, exclusion scope), the audit report’s status was updated, and the caliber impact was measured rather than assumed: the static dashboard is untouched (dashboards are generated artifacts, not live queries), and the next regeneration will see the batch pool shrink from $\langle 119859 \rangle$ to $\langle 119858 \rangle$ files with $\langle 10 \rangle$ template theorems removed from both numerator and denominator of the batch caliber — the intended effect of quarantining a contamination source.

3.5 Lifecycle case study: the Chern–Simons chain

The strongest single narrative in the repository is the staged CLAIM→THEOREM upgrade along the Chern–Simons dependency chain:

1. **Claim identification:** the fine-structure coincidence $\alpha^{-1} \approx n_{CS} = 137$ is registered as a CLAIM with an explicit falsifiability criterion (first-principles prediction of α^{-1} to ± 0.01 , independent of loop-integer self-tuning).
2. **Chain census:** the 8-file Chern–Simons side-chain was inventoried with per-file placeholder/axiom census, exposing exactly which links of the purported derivation were load-bearing debt.
3. **Deepest provable link upgraded:** the stratified Chern-number identity in the integer-weight sector (`stratifiedChernNumber_int_weights`) was formalized via mathlib delegation (`Int.cast_sum`, `Int.cast_mul`), independently compiled (exit 0, $\langle 6.7 \rangle$ s), with `#print axioms` reduced to the standard trio.

4. **Honest labeling:** the upgraded statement is labeled what it is — a $\mathbb{Z}$ -ring-closure identity in the integer-weight sector, *not* the topological integrality of the real-weight Chern–Simons functional; that stronger statement remains open, and the remaining gap chain (Chern–Weil theory absent from mathlib; the spectral-action bridge axiom; the level placeholder $\equiv 137$) is registered alongside it.
5. **Registration:** the tier assignment and its evidence live in the central registry (`proof_status.md` sec 3), not in the paper draft — the draft cites the registry.

The Dedekind unique-factorization sub-case demonstrates the other half of the discipline: a delegation wrapper around mathlib’s `normalized_factor_uniq` proves a THEOREM with *zero new mathematics*, and the registry says so explicitly, rather than letting the tier label imply novelty.

3.6 Conflict inventory (C1–C6)

The paper’s own numbers are governed by the same discipline it advocates. Six inter-document caliber conflicts were inventoried by the internal review; their resolution landed with the P0 reconciliation (`framework/p0_reconciliation_report.md`, 2026-08-21), which this skeleton now cites as its canonical-number source:

ID	Subject	Status (P0 landed)
C1	axiom total (478 vs 448 vs “ $\sim 350+$ ”)	resolved: 253 live-lean $\equiv$ registry residual; 448 = registry baseline caliber
C2	Agda postulates (“ ~ 149 ” vs 55 measured)	resolved: 7 files / 131 name-level postulates (149 = planning value, 55 = string-caliber)
C3	proof rate (3.91% vs 99.79%)	resolved: 3.91% authoritative; legacy dashboard marked stale (Sec. 3.2)
C4	batch sorry caliber	resolved (string-caliber false positive; Sec. 3.3)
C5	sweep3 internal ledger (37 vs 31)	resolved: 31; method-table 37 was a miscount (corrected in sweep3 report)
C6	“136 redeemed” composition	resolved: canonical 196 = five-class decomposition (Sec. 3.1); “136/198” were double-count calibers

Every `<...>` in this paper is therefore a declared caliber, traceable to a source file; C6-related figures have been replaced by P0 canonical numbers, and the remaining values are verified v7.98 baselines — the same (number, caliber) discipline of Sec. 2.3, applied to the paper itself.

4 Discussion

4.1 Caliber discipline as the central thesis

The paper’s transferable claim is not the specific numbers of one library but a methodological proposition: *in a formal library that mixes human and machine authorship, a proof-progress metric is only meaningful as a (number, caliber) pair, and the caliber must be declared, machine-checkable, and enforced by the same CI that enforces the rest of the honesty layer.* The two

self-caught incidents are the evidence that this is load-bearing rather than ceremonial: the [\(30.2%\)](#) contamination alarm (F3) was produced by a caliber mismatch that any of the library’s own automated gates could have made silently permanent had the false positive not been audited to closure; the 25× dashboard disagreement (F2/C3) survived in production documents for days precisely because both files carried the same informal label (“honest proof rate”) with different implicit calibers. A reviewer or meta-scientist who accepts only one claim from this paper should accept this one.

4.2 Related work

- **mathlib #print axioms discipline and linter ecosystem:** the community practice of listing axiom dependencies of shipped theorems, and linters that police unwanted `sorry` / `axiom` declarations. Delta: we manage a taxonomy of *legitimate* axiom classes plus a redemption ledger, not only a prohibition regime.
- **Isabelle/HOL conservative-extension and import audits:** AFP’s review process and Isabelle’s emphasis on conservative extensions. Delta: our G1–G3 transformations are claimed to be conservative at the governance level; formalizing that claim (Sec. 4.5) would connect the two traditions.
- **Metamath and minimal-axiom-economy traditions:** explicit bookkeeping of assumed primitives. Delta: our problem is not economy but *debt retirement under scale*.
- **LLM proof-integrity literature (2024–2026):** documented incidents of fabricated or silently weakened proofs in machine-generated corpora. Delta: we contribute a full-population contamination audit that reversed a string-caliber alarm, plus a quarantine protocol with measured (not asserted) caliber impact.

4.3 Limitations

- **Numbers are not yet canonical.** Six caliber conflicts (C1–C6) are under P0 reconciliation; every number here is a declared v7.98 baseline. The paper is un-submittable in this state by design — the skeleton exists so that the writing and the number reconciliation can proceed in parallel without the former quietly hard-coding the latter.
- **No containerized reproducibility artifact.** Compile logs exist and are three-way consistent (lake-manifest vs. two independent builds), but carry local paths; a Docker/CI artifact package is P1-8.
- **Metatheorems missing.** The conservativity of G1–G3 transformations is asserted at the governance level, not formalized.
- **Single-library case study.** The taxonomy was induced from one library; cross-library validation (mathlib, AFP, or another AI-generated corpus) is future work.
- **Compile-log provenance.** Independent compilation was performed on one machine class; the [\(6.7\)](#)s compile time is an environment-dependent anecdote, not a benchmark.

4.4 Anticipated objections

1. “*AI-generated content cannot be trusted.*” The paper’s position: precisely — which is why every artifact class carries a registration-and-audit trail, and why the two metric defects were caught by the loop itself. The claim is not that the library is trustworthy; it is that its trust posture is *explicit*.

2. “*Your axioms are non-standard.*” The taxonomy exists to separate genuine primitives from debt; the debt is being retired with graded honesty; the residual is declared, not hidden.
3. “*The batch corpus is contamination.*” Full-population audit: code-level contamination is exactly one file, now quarantined; the other ⟨36198⟩ string hits are denials, and the corpus’s triviality is a *labeled property* of the corpus, not a hidden one.
4. “*Your headline numbers drifted by 25×.*” Yes — and the drift is the finding: undecalared calibers, not dishonest arithmetic. The conflict inventory and the placeholder discipline in this very draft are the corrective.

4.5 Future work

Three lines, in dependency order: (i) **metatheorems** — formalize conservativity of the G1–G3 transformation classes (the infrastructure target is a Lean proof that each transformation extends the ambient theory conservatively); (ii) **TrivialBench sampling under the code caliber** — the revised A-2 clause ships with this program, and its first full sampling round will exercise the quarantine list; (iii) **reproducibility package** — containerized builds publishing `#print axioms` artifacts per release. Longer term, the registry + redemption + caliber-discipline stack is library-agnostic; porting it to an external corpus is the natural validation study.

4.6 Ethics and transparency

The library and this paper were produced with AI assistance under human tasking and review. The repository’s transparency conventions (AI-assistance markers on every artifact, deleted-claim ledger, no-silent-deletion rule, CI-enforced honesty markers) apply to this manuscript: the draft status, baseline numbers, and pending reconciliation are visible by construction, and the supplementary material includes the build log of the exact $\text{\LaTeX}$ compile that produced the accompanying PDF.

References